

Operation Threshold

Debt-Burden Risk & First-Payment Default — Executive Summary & Recommendations

The question

“We’re approving loans on credit score and income verification, but nobody has checked whether size-of-loan-versus-income is quietly doing more damage than either. I don’t want a guess. I want a number.”

— Head of Credit Risk

Debt burden — how large a loan is relative to the borrower’s income — turns out to be the strongest predictor of default in this portfolio, well ahead of most of what the initial sample review suspected. A few other suspected patterns (first-payment timing, officer inconsistency) are real but weaker than they first looked, and one doesn’t hold up at all. This summary says plainly which is which.

What the data shows

- **Debt burden predicts default.** Default rate rises roughly 8x from Low DTI (5.48%) to Severe DTI (43.50%), and this number reverse-engineers correctly against the portfolio’s total defaults — this is the finding to act on. *(strong)*
- **A missed first payment rarely means default.** 63% of loans that missed their first payment went on to repay normally with no default at all (default = 3 or more missed payments). A single missed payment is a weak warning sign by itself. *(strong)*
- **Good credit score stops protecting borrowers once debt burden is severe.** At Severe DTI, Fair, Poor, and Unknown credit tiers all converge to roughly the same 43–45% default rate — credit score stops discriminating risk once the loan itself is oversized. *(moderate)*
- **Defaults happen gradually more often than immediately, at every debt-burden level.** 75–80% of defaults are gradual regardless of DTI band. Higher DTI only slightly raises the chance of immediate failure — real, but a small effect, not the dramatic pattern first suspected. *(moderate)*
- **SME loans and two branches (Manchester, Newcastle) default more than the rest of the portfolio.** Worth a closer look, though a smaller effect than the DTI-band finding above. *(moderate)*
- **The loan-officer “ranking” isn’t real.** Of 40 officers, only 3 have a default pattern that’s statistically different from the portfolio average once sample size is accounted for. The rest of the spread you’d see in a sorted list is normal statistical noise, not a skill difference — don’t act on it as a ranking. *(weak — not supported)*

Recommendations

- **Tighten approval criteria above ~35% DTI** (the High/Severe boundary) — this is where default risk and money at risk both concentrate. Owner: Underwriting. *(action)*
- **Treat debt-to-income, not credit score alone,** as the primary risk signal once a loan is large relative to income. Owner: Underwriting. *(action)*
- **Give SME applications and the Manchester/Newcastle branches** extra review. Owner: Portfolio risk. *(action)*
- **Use a missed first payment as a prompt for early borrower contact,** not as grounds to flag the loan as high-risk. Owner: Servicing / collections. *(action)*
- **Don’t rank or act on individual loan officers from this data.** Only 3 of 40 are statistically distinguishable from average. Owner: Risk Committee. *(action)*

A note on the confidence labels

Strong: the pattern holds up against a cross-check and is large enough that it’s very unlikely to be chance. Moderate: the pattern is real, but the effect is small or part of the underlying data was too thin to fully confirm it. Weak / not supported: the pattern looks meaningful at first glance, but a proper statistical check shows it’s within the range you’d expect from randomness alone — treat it as a lesson in what not to act on, not as a discovery.